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23A2244401

II B. Tech. II Semester Supplementary Examinations, Nov/Dec-2025

DATA ENGINEERING

(CSE(DS))

Time: 3 Hours**Dt: 29-11-2025****Max. Marks: 70**

Question paper contains two parts A and B.

Part A is compulsory

Part B consists of 5 Units. Answer any one full question from each unit.

PART A		10X2M=20M	
Q.No.	Questions	Marks	BTL
1. a)	How does Data Engineering differ from Data Science?	2M	1
b)	Explain the importance of Data Maturity in an organization?	2M	2
c)	What are Source Systems in the Data Engineering Life Cycle?	2M	1
d)	Explain the role of Data Transformation in the Data Engineering Life Cycle?	2M	2
e)	What are the major concepts in Data Architecture?	2M	1
f)	What types of data are classified as unstructured data?	2M	1
g)	List the primary difference between a Data Lake and a Data Lakehouse?	2M	2
h)	Identify the primary difference between Batch Ingestion and Stream Ingestion?	2M	3
i)	Define Data Modeling in the context of data engineering.	2M	1
j)	What is Reverse ETL, and how is it different from traditional ETL?	2M	1

PART B		5x10M=50M	
Q.No.	Questions	Marks	BTL
UNIT I			
2. a)	Compare and contrast the roles of Data Engineers and Data Scientists in an organization.	5M	2
b)	Discuss the business responsibilities of a Data Engineer and their impact on business decision-making.	5M	4
OR			
3. a)	Describe the evolution of Data Engineers and how their roles have transformed over time.	5M	2
b)	Explain the role of Data Engineers in ensuring data quality and reliability within an organization.	5M	2
UNIT II			
4. a)	Describe the key responsibilities of Data Operations in the Data Engineering Life Cycle.	5M	2
b)	Describe how the integration of security measures in the Data Engineering Life Cycle helps mitigate risks and ensures compliance.	5M	2

OR			
5. a)	Discuss the challenges involved in managing Data Architecture in large-scale data systems and how they can be overcome.	5M	2
b)	Discuss the importance of Data Security in the Data Engineering Life Cycle and its impact on data integrity and privacy.	5M	2
UNIT III			
6. a)	Discuss the difference between Enterprise Architecture and Data Architecture and explain how they align to achieve business objectives.	5M	3
b)	Describe the major architecture concepts in designing a Data Architecture and how they impact the scalability and flexibility of data systems.	5M	2
OR			
7. a)	Illustrate the process of designing good Data Architecture by outlining the key considerations and best practices for building a robust data infrastructure.	5M	2
b)	Explain the concept of Change Data Capture (CDC) and its relevance to tracking changes in source systems in real-time.	5M	2
UNIT IV			
8. a)	What are the key advantages and challenges of using Data Lakes for storing large volumes of unstructured data?	5M	1
b)	Discuss the key factors to consider when implementing a data ingestion system for a large-scale enterprise.	5M	3
OR			
9. a)	Discuss how Data Warehouses are optimized for analytical processing and how they differ from other storage systems in terms of data structure and query performance.	5M	3
b)	Explain the role of metadata management during the ingestion phase and its importance in ensuring data consistency and quality.	5M	2
UNIT V			
10. a)	Discuss the life cycle of a query in a data processing pipeline, from initiation to execution, and explain the role of the query optimizer in this process.	5M	3
b)	Compare Business Analytics and Operational Analytics, highlighting their use cases, tools, and impact on business decision-making.	5M	2
OR			
11. a)	Elaborate on the concept of transformations in data engineering, and how streaming transformations are different from traditional batch transformations.	5M	2
b)	How does Reverse ETL enable the synchronization of data between a data warehouse and business applications? Discuss the tools and techniques involved in this process.	5M	2

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